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A Broader Impacts

This work was motivated by the goal of training more safe and trustworthy AI systems. We believe that LAT will be practically useful for training better models. However, we emphasize that LAT is a value-neutral technique for training AI systems to align with their developer’s goals. It is important not to conflate AI alignment with safety (Khlaaf, 2023). We believe that this work will contribute to helpful progress, but we emphasize that many of the risks from AI systems come from misuse and adverse systemic effects as opposed to unintended hazards such as the ones we work to address.

B Loss Functions for LAT

B.1 RT-EAT-LAT

Here, we describe the RT-EAT-LAT method described in Section 4.1 in greater detail. We assume we are given two datasets - a dataset of harmful requests and *pairs* of preferred and rejected completions $\mathcal{D}_p = \{(x_i, c_i, r_i)\}$, and a generic dataset of **benign** requests and helpful completions $\mathcal{D}_b = \{(x_i, y_i)\}$. For each batch, we train the adversarial attack δ to minimize $\mathcal{L}_{\text{attack}}$:

$$\mathcal{L}_{\text{attack}} = \underbrace{-\log P(r_i | g_\theta(f_\theta(x_i) + \delta_i))}_{\text{Move towards harmful completions}} + \underbrace{-\log(1 - P(c_i | g_\theta(f_\theta(x_i) + \delta_i)))}_{\text{Move away from harmless completions}} \quad (3)$$

We additionally add the constraint that $\|\delta_i\|_2 \leq \epsilon$, where ϵ is a hyperparameter, to restrict the adversary’s power. We then train the model parameters θ against these adversarial attacks by minimizing $\mathcal{L}_{\text{model}}$. We define $\mathcal{L}_{\text{model}}$ in terms of the loss functions $\mathcal{L}_{\text{defense}}$ and $\mathcal{L}_{\text{benign}}$:

$$\mathcal{L}_{\text{defense}} = \sum_{(x_i, c_i, r_i) \sim \mathcal{D}_p} \underbrace{-\log P(c_i | g_\theta(f_\theta(x_i) + \delta_i))}_{\text{Move towards harmless completions}} + \underbrace{-\log(1 - P(r_i | g_\theta(f_\theta(x_i) + \delta_i)))}_{\text{Move away from harmful completions}} \quad (4)$$

$$\mathcal{L}_{\text{model}} = \mathcal{L}_{\text{defense}} + \mathcal{L}_{\text{benign}} \quad (5)$$

We can use one of two different benign loss terms:

$$\mathcal{L}_{\text{benign, SFT}} = \sum_{(x_i, y_i) \sim \mathcal{D}_b} -\log P(y_i | g_\theta(f_\theta(x_i))) \quad (6)$$

$$\mathcal{L}_{\text{benign, KL}} = \sum_{(x_i, y_i) \sim \mathcal{D}_b} \text{KL} [P(y_i | g_{\theta^*}(f_{\theta^*}(x_i))) \parallel P(y_i | g_\theta(f_\theta(x_i)))] \quad (7)$$

where θ^* are the weights of the frozen reference model. Note that $\mathcal{L}_{\text{benign}}$ is always calculated on inputs where no adversarial attack is present.